MSG No. 184.194 $P6'cc'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `6'..`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,2,3,7,8,9]`
2	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	`[4,5,6,10,11,12]`

Table 2: Wyckoff site: `4b`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$	`[1,2,3]`
2	$\left[\frac{1}{3},\, \frac{2}{3},\, z+\frac{1}{2}\right]$	`[4,5,6]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, z\right]$	`[7,8,9]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, z+\frac{1}{2}\right]$	`[10,11,12]`

Table 3: Wyckoff site: `6c`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, z\right]$	`[1,8]`
2	$\left[0,\, \frac{1}{2},\, z\right]$	`[2,9]`
3	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	`[3,7]`
4	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	`[4,11]`
5	$\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[5,12]`
6	$\left[\frac{1}{2},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[6,10]`

Table 4: Wyckoff site: `12d`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x-y,\, z]$	`[2]`
3	$[-x+y,\, -x,\, z]$	`[3]`
4	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	`[4]`
5	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	`[5]`
6	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	`[6]`
7	$[x-y,\, x,\, z]$	`[7]`
8	$[-x,\, -y,\, z]$	`[8]`
9	$[y,\, -x+y,\, z]$	`[9]`
10	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	`[10]`
11	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	`[11]`
12	$\left[y,\, x,\, z+\frac{1}{2}\right]$	`[12]`